

# PRIYESH BHALALA

## AI / ML Engineer

Vadodara, Gujarat | M.Sc. Artificial Intelligence, Parul University

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### TECHNICAL SKILLS

**Languages:** Python, C

**ML & Deep Learning:** TensorFlow, Keras, MobileNetV2, CNN, Transfer Learning, Fine-Tuning, scikit-learn, OpenCV, Transformers (GPT, BERT, LLaMA, Qwen), GANs, VAEs, Diffusion Models Data Augmentation, Label Smoothing, Dropout, Confusion Matrix, Classification Report, K-Means Clustering, RFM Analysis, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Bias Mitigation

**Data & Visualization:** pandas, NumPy, Matplotlib, Seaborn, Feature Engineering

**Web & Frameworks:** Django, Flask, HTML, CSS, JavaScript

**Databases:** MySQL, MongoDB, Oracle SQL — joins, subqueries, PIVOT/UNPIVOT, CASE

**Scraping & Automation:** BeautifulSoup, Playwright, Selenium

**Tools & Platforms:** Power BI, Jupyter Notebook, VS Code, Git, GitHub

### WORK EXPERIENCE

#### Python Developer | Innovativecode IT Solutions

Jan 2024 – Dec 2024

- Built web scraping and data extraction scripts using BeautifulSoup and Playwright to collect structured data from multiple target websites.
- Automated data collection workflows, removing manual steps from the pipeline and reducing errors in downstream data outputs.
- Optimized scraper performance by tuning request handling, retry logic, and HTML parsing strategies to improve speed and data accuracy.
- Worked with team members to debug, improve, and deploy Python-based automation tools across multiple business use cases.

### PROJECTS

#### Fabric Classification Using Deep Learning — Python, TensorFlow, Keras, MobileNetV2, PIL, Matplotlib, Seaborn

- Built an image classification system using MobileNetV2 (CNN with Transfer Learning) to automatically identify 4 fabric types — cotton, polyester, silk, and wool — from photos, replacing error-prone manual inspection.
- Preprocessed all fabric images by resizing to 224x224 pixels and applied data augmentation (rotation, flipping, zoom) to increase training set variation and improve generalization.
- Trained in two phases: froze MobileNetV2 base layers first to train the classifier head, then unfroze the last 40 layers for fine-tuning — improving accuracy on visually similar fabric classes.
- Used label smoothing and dropout regularization to reduce overfitting; evaluated model performance using a confusion matrix, classification report (Precision, Recall, F1-Score per class), and accuracy/loss curves.
- Built a prediction module that accepts a fabric image and returns the predicted class with a confidence score — designed for use by quality checkers in textile factories and e-commerce catalog managers.

#### Customer Segmentation for E-Commerce Retail — Python, scikit-learn, pandas, NumPy, SQL

- Built a customer segmentation model using K-Means Clustering to group customers by purchasing behavior and identify distinct customer types from transactional data.
- Engineered RFM (Recency, Frequency, Monetary) features from raw transaction records to quantify customer value and produce actionable segments for marketing decisions.
- Cleaned and aggregated transactional data using pandas and SQL — including joins, subqueries, and GROUP BY — to build a structured feature table as model input.

- Segment results used to guide loyalty program design, targeted marketing strategies, and product recommendation logic.

**Health Management Analysis Dashboard** — *Power BI, Oracle SQL*

- Built an interactive Power BI dashboard connected to Oracle SQL to visualize patient records, medical trends, and operational data for healthcare analysis.
- Gathered stakeholder requirements and user stories to design dashboard layouts and select appropriate chart types for each data category.
- Wrote SQL queries using inner, outer, and cross joins, subqueries, CASE statements, and PIVOT/UNPIVOT to build the data model powering the dashboard.
- Diagnosed and resolved database issues including performance bottlenecks, data inconsistencies, and errors affecting report output accuracy.

**EDUCATION**

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**Master of Science — Artificial Intelligence | Parul University**  
*Vadodara, Gujarat*

*Sept 2024 – May 2026*

**Bachelor of Computer Application | Marwadi University**  
*Rajkot, Gujarat*

*Sept 2021 – April 2024*